

Module 9: Inheritance Genetics - Practice Quiz

Use this low-stakes quiz after reviewing the module keys and learning questions.

1. An organism's genotype is its:

- A. set of observable physical traits
- B. surrounding environment
- C. combination of alleles
- D. total number of cells
- Answer: C
- Why: Genotype is the combination of alleles an organism carries. Phenotype is the observable trait that results.

2. Two heterozygotes are crossed ($Aa \times Aa$). What fraction of offspring are expected to show the recessive trait?

- A. $1/4$
- B. $1/2$
- C. $3/4$
- D. 0
- Answer: A
- Why: $Aa \times Aa$ gives a $1 AA : 2 Aa : 1 aa$ ratio. Only the aa offspring ($1/4$) show the recessive trait.

3. A heterozygote (Aa) shows the dominant trait because:

- A. it carries no recessive allele
- B. recessive alleles are destroyed
- C. both alleles are always expressed equally
- D. one dominant allele is enough to mask the recessive allele
- Answer: D
- Why: In simple dominance, one dominant allele determines the phenotype. The recessive allele is present but masked.

4. What is a pedigree used for?

- A. measuring the size of a cell
- B. tracing how a trait passes through a family to infer its inheritance
- C. editing genes directly
- D. counting chromosomes in soil samples
- Answer: B
- Why: A pedigree diagrams family relationships and phenotypes so you can reason about how a trait is inherited.